DATA WAREHOUSING ACTIVITIES

Activities 1 to 4: Solutions

Name: Manoj A

Register Number: 192421295

Activity 1: Design a Star Schema

Question: *Design a star schema for a college data warehouse that analyses student academic performance across departments, courses and semesters.*

Design approach

The business event to analyse is a student's result in one course during one semester, so the fact table holds one row at exactly that grain. The numeric results (marks, grade point, attendance) are the measures. Who, what, where and when are described by four dimension tables placed around the fact table, which gives the star shape.

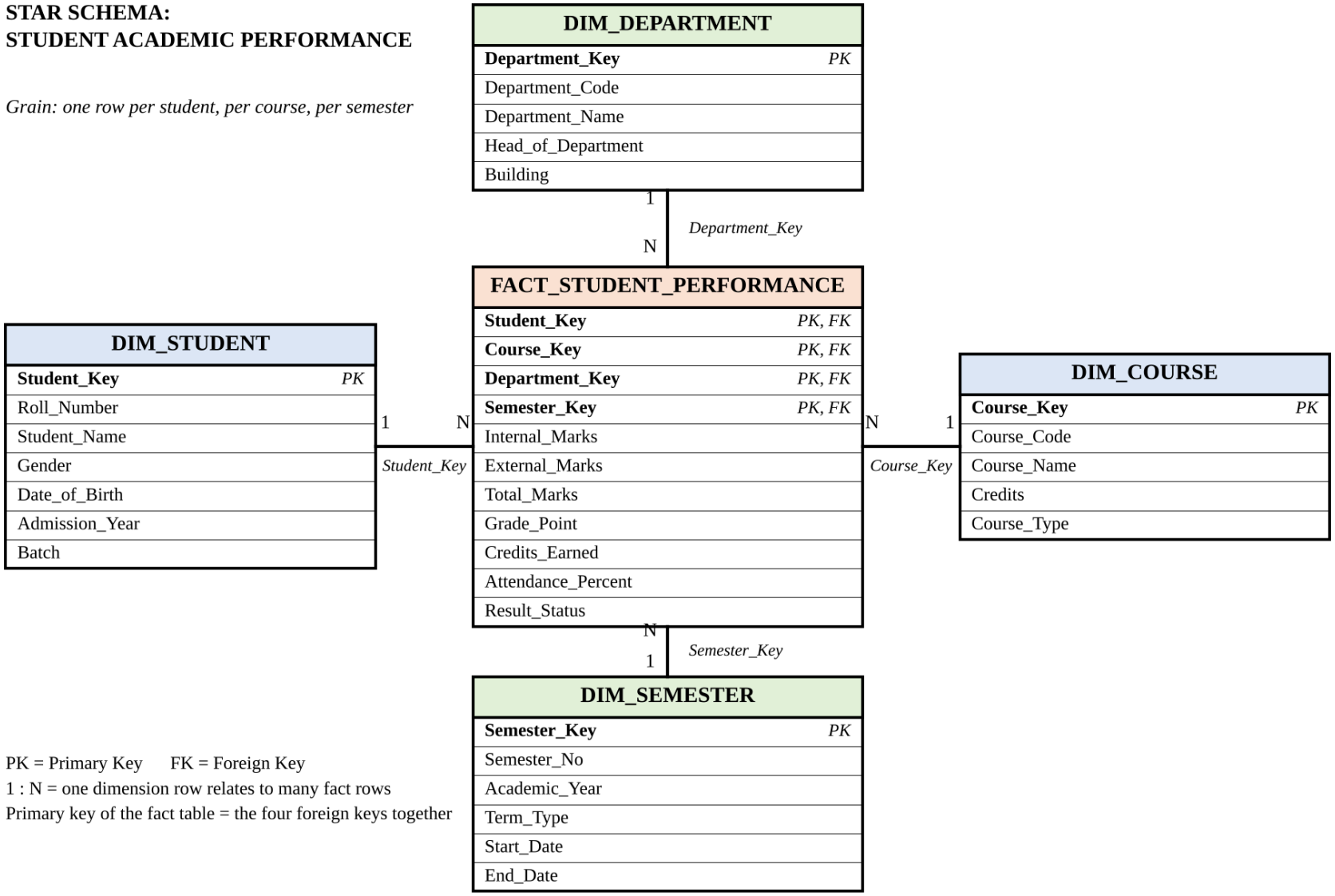
Table 1: Fact and dimension tables with keys

Table	Attributes	Primary Key	Foreign Keys
FACT_STUDENT_PERFORMANCE	Student_Key, Course_Key, Department_Key, Semester_Key, Internal_Marks, External_Marks, Total_Marks, Grade_Point, Credits_Earned, Attendance_Percent, Result_Status	Composite of the four foreign keys	Student_Key, Course_Key, Department_Key, Semester_Key
DIM_STUDENT	Student_Key, Roll_Number, Student_Name, Gender, Date_of_Birth, Admission_Year, Batch	Student_Key	None

Table	Attributes	Primary Key	Foreign Keys
DIM_COURSE	Course_Key, Course_Code, Course_Name, Credits, Course_Type	Course_Key	None
DIM_DEPARTMENT	Department_Key, Department_Code, Department_Name, Head_of_Department, Building	Department_Key	None
DIM_SEMESTER	Semester_Key, Semester_No, Academic_Year, Term_Type, Start_Date, End_Date	Semester_Key	None

Each dimension uses a surrogate key as its primary key. The fact table repeats those four keys as foreign keys, and together they identify a fact row uniquely. The star schema diagram is shown on the next page. A typical question it answers is the average grade point by department and semester.

Figure 1: Star Schema for Student Academic Performance



Activity 2: Perform ETL Operations

Question: *Integrate student data from the admission, examination and attendance systems into the warehouse by Extract, Transform and Load.*

Step 1: Extract

Data is copied unchanged from the three source systems into the staging area. The sample records below contain the problems named in the question.

Table 2: Admission system (input)

Stu_ID	Name	Department	Admission_Date
A101	Ravi Kumar	C.S.E	12-06-2023
A102	Priya S	Computer Science	2023/07/05
A102	Priya S	Comp. Science	05-07-2023
A103	Arun M	ECE	(blank)
A104	Meena R	Electronics & Comm	June 14, 2023

Table 3: Examination system (input)

Stu_ID	Marks
A101	78
A102	85
A103	(blank)
A104	66

Table 4: Attendance system (input)

Stu_ID	Attendance_%
A101	92
A102	88
A103	75
A104	(blank)

Step 2: Transform

The extracted records are cleaned and combined using the rules below.

Table 5: Transformation rules applied

Problem	Rule applied	Effect on sample data
Duplicate records	Keep one record per Student_ID	A102 appeared twice and is reduced to one row
Inconsistent department names	Map every variant to a standard code using a lookup table	C.S.E, Computer Science, Comp. Science become CSE; ECE, Electronics & Comm become ECE
Different date formats	Convert all dates to YYYY-MM-DD	12-06-2023 becomes 2023-06-12; June 14, 2023 becomes 2023-06-14
Missing admission date	Insert default 1900-01-01 and flag the row	A103 gets 1900-01-01
Missing marks	Keep NULL (not zero) and flag as pending	A103 marks stay NULL
Missing attendance	Fill with the average of available values: (92 + 88 + 75) / 3	A104 attendance becomes 85.0
Integration	Join admission, examination and attendance on Student_ID	One row per student

Step 3: Load

The cleaned, integrated rows are loaded into the target student table of the warehouse. The first run is a full load; later runs add only new or changed records (incremental load). The final loaded data is shown below.

Table 6: Target table after loading

Student ID	Name	Dept	Admission Date	Marks	Attendance %	Quality Flag
A101	Ravi Kumar	CSE	2023-06-12	78	92.0	Clean
A102	Priya S	CSE	2023-07-05	85	88.0	Duplicate removed
A103	Arun M	ECE	1900-01-01	NULL	75.0	Date defaulted, marks pending
A104	Meena R	ECE	2023-06-14	66	85.0	Attendance imputed

Activity 3: Demonstrate OLAP Operations

Question: *Using a sales warehouse with Time, Product and Location dimensions, demonstrate roll-up, drill-down, slice, dice and pivot.*

Sample dataset

The measure is Sales Amount in thousands of rupees. Every cell is one fact record identified by a product, a location and a month.

Table 7: Base sales data (Sales Amount, Rs. thousand)

Product	Location	2024-Jan	2024-Feb	2025-Jan	2025-Feb
Laptop	Chennai	120	135	150	160
Laptop	Mumbai	140	130	165	155
Laptop	Delhi	110	125	130	145
Phone	Chennai	200	210	230	240
Phone	Mumbai	220	215	250	260
Phone	Delhi	180	190	205	215
Tablet	Chennai	80	85	90	95
Tablet	Mumbai	70	75	85	90
Tablet	Delhi	60	65	70	80

(a) Roll-up: monthly sales to yearly sales

Roll-up climbs the Time hierarchy from Month to Year, so the two monthly values of each year are added together.

Table 8: Yearly sales by product and location

Year	Product	Chennai	Mumbai	Delhi	Total
2024	Laptop	255	270	235	760
2024	Phone	410	435	370	1215
2024	Tablet	165	145	125	435
2025	Laptop	310	320	275	905
2025	Phone	470	510	420	1400
2025	Tablet	185	175	150	510

(b) Drill-down: yearly sales to month-wise sales

Drill-down is the reverse step. Taking the 2025 yearly figures from Table 8, the same data is shown month by month.

Table 9: Month-wise sales for 2025

Month	Product	Chennai	Mumbai	Delhi	Total
Jan 2025	Laptop	150	165	130	445
Jan 2025	Phone	230	250	205	685
Jan 2025	Tablet	90	85	70	245
Feb 2025	Laptop	160	155	145	460
Feb 2025	Phone	240	260	215	715
Feb 2025	Tablet	95	90	80	265

(c) Slice: one specific product

Slice fixes one dimension value, Product = Laptop, and keeps the remaining dimensions.

Table 10: Slice for Product = Laptop

Month	Chennai	Mumbai	Delhi	Total
2024-Jan	120	140	110	370
2024-Feb	135	130	125	390
2025-Jan	150	165	130	445
2025-Feb	160	155	145	460

(d) Dice: selected products, locations and periods

Dice selects values on several dimensions at once: Product in {Laptop, Phone}, Location in {Chennai, Mumbai} and Time in {2025-Jan, 2025-Feb}.

Table 11: Dice result

Month	Product	Chennai	Mumbai	Total
2025-Jan	Laptop	150	165	315
2025-Jan	Phone	230	250	480
2025-Feb	Laptop	160	155	315
2025-Feb	Phone	240	260	500

(e) Pivot: rotate rows and columns

Pivot turns the view around without changing the data. Table 12 compares 2025 sales with products as rows and locations as columns; Table 13 rotates it so locations become rows and products become columns.

Table 12: 2025 sales, products as rows

Product	Chennai	Mumbai	Delhi	Total
Laptop	310	320	275	905
Phone	470	510	420	1400
Tablet	185	175	150	510
Total	965	1005	845	2815

Table 13: 2025 sales after pivot, locations as rows

Location	Laptop	Phone	Tablet	Total
Chennai	310	470	185	965
Mumbai	320	510	175	1005
Delhi	275	420	150	845
Total	905	1400	510	2815

Activity 4: Data Warehouse Architecture for a Hospital

Question: *Design an architecture that integrates patient registration, laboratory, pharmacy and billing data for management decision-making.*

The architecture is drawn in Figure 2 on the next page. Data flows from left to right through seven components, and the function of each is given below.

Table 14: Components and their functions

Component	Function
Data sources	Operational systems for patient registration, laboratory, pharmacy and billing that generate the daily records.
ETL process	Extracts data from the sources, cleans and standardises it (for example one patient ID format) and loads it forward.

Component	Function
Staging area	A temporary workspace where extracted data is validated and de-duplicated before it touches the warehouse.
Central data warehouse	Stores integrated, historical hospital data organised by subject, with a metadata repository describing every table and field.
Data marts	Smaller subject-specific stores (clinical, pharmacy, finance) that give each department fast access to only its own data.
OLAP tools	Let users explore the marts from many angles using roll-up, drill-down, slice, dice and pivot.
Reporting tools	Present results as standard reports, dashboards with KPIs and ad-hoc queries for decision makers.

How the architecture supports decision-making

Because all four systems feed one warehouse, management sees a single, consistent version of the hospital's data instead of four separate reports. The history stored in the warehouse allows trends to be studied, such as monthly admissions, medicine consumption or revenue per department. The data marts return answers quickly to each department, and the OLAP and reporting tools let managers move from a summary figure to its detail without technical help. This supports decisions on staffing, stock of medicines, bed usage and billing follow-up.

Figure 2: Data Warehouse Architecture of the Hospital

